SIEGFRIED BOZZA

Full-Stack Web3 Engineer · TypeScript, Solidity, Node.js, Next.js

Remote, location-independent · Annecy, France (CET) · siegfried.bozza@protonmail.com · GitHub · Portfolio · LinkedIn

PROFESSIONAL SUMMARY

Full-Stack Web3 Engineer who builds the off-chain systems that on-chain products depend on: event pipelines, wallet and settlement backends, durable AI-agent workflows, and the dApps on top. TypeScript since 2022, in production since 2024 on a Next.js / Supabase SaaS with CQRS and hexagonal architecture; Solidity and Foundry with Chainlink integrations on Ethereum and Base testnets. Former molecular-biology researcher (three peer-reviewed papers), which drives my DeSci work. Every project ships with tests, written design decisions, and its known limitations.

TECHNICAL SKILLS

Web3 & Smart Contracts: Solidity, Foundry (contracts, tests, deploy scripts), OpenZeppelin (ERC-20/721/1155), Chainlink (VRF v2.5, Automation, Price Feeds), Ethereum, Base, Viem, Wagmi, EIP-712, EIP-1559, HD wallets (BIP-32/39/44), Multicall3, OriginTrail DKG, Reown AppKit, IPFS, Alchemy, Tenderly

Testing & DevOps: Vitest, Cypress, Jest, GitHub Actions, Vercel, CI/CD, Git Flow

Backend & Data: Node.js, Hono, Supabase (RPC, RLS, Realtime, Edge Functions, CRON), Neon Postgres, Drizzle ORM, SQLite, CQRS, event-driven design, idempotent workflows, Stripe API

Frontend & UI: React 19, Next.js (App Router), TypeScript, TanStack Query / Table, nuqs, Tailwind CSS, shadcn/ui, Zod, Recharts, React Hook Form, Lingui

AI & Orchestration: LangGraph.js (agentic graphs, HITL, checkpointers), Inngest (durable execution), Gemini, Exa AI

Languages: French (native) · English (fluent)

WORK EXPERIENCE

Prismo.io — Hybrid, France · Full-Stack Engineer · Aug 2024 – Present

- Built real-time Next.js dashboard components with TanStack Table, nuqs, Recharts, shadcn/ui, Tailwind CSS, React Suspense and Lingui, bringing table, filter and chart interactions into one interface.
- Implemented CQRS read/write separation with Supabase triggers, RPC functions and a typed client, and hexagonal architecture (ports and adapters) that keeps domain logic testable and independent of infrastructure.
- Added Vitest unit tests and Cypress end-to-end tests for critical workflows; CI requires them to pass before a merge to main.
- Delivered in two-week sprints via Linear with peer review on every PR and Git Flow; AI agents (Cursor) for planning and implementation, with their output reviewed like any other PR.

Independent study & open source — France & Indonesia · Jul 2023 – Aug 2024

- Studied full-stack and Web3 development while travelling, building my own React and Next.js projects (Tailwind CSS, shadcn/ui, MongoDB, Jest).
- Contributed front-end features to DeXter-on-Radix, an open-source order-book DEX: CSV order export, a stats widget, mobile navigation and a DaisyUI v4 upgrade.

Renuo AG — Zurich, Switzerland · Software Engineer Intern · Jul 2022 – Jun 2023

- Built and shipped full-stack features using Ruby on Rails, JavaScript/TypeScript, React, and RSpec in agile delivery cycles, supporting production releases across multiple agency client accounts.

SELECTED PROJECTS & OPEN SOURCE

VeriSci — Open Ratings for Life-Science Papers · Oct 2026 · GitHub · Article · Live

- Papers go to the OriginTrail DKG as EIP-712-signed Knowledge Assets; anyone can request a rating on Base Sepolia. Gemini answers a reporting checklist (CONSORT, ARRIVE…) with quotes that code checks.
- Every write converges: stores, mints, fulfils and cancels read state first. HMAC-verified Alchemy webhooks, request-derived Inngest event ids, a cron reconciler and a serialized oracle lane.
- 35 ADRs, 1,714 Vitest and 74 Foundry tests (fuzz + invariants); staging and production isolated. On testnet and under construction: human review with Chainlink VRF is next.
- Built in three weeks alongside my full-time job, with Claude Code in a harness I set up.

Wallet Watcher — Custody Backend Slice · Sep – Oct 2026 · GitHub · Article

- Node.js / TypeScript service on Ethereum Sepolia (viem, Hono, Zod, SQLite): BIP-39/44 HD wallets, an append-only ETH ledger and signed EIP-1559 withdrawals behind a localhost admin API.
- Outflows booked exactly once from receipts, bigint wei end to end, Multicall3 balance snapshots pinned to their block, a worst-case fee check before signing, and key isolation (only the signer derives keys); 156 offline tests in CI.

BioVerify — Agentic DeSci DApp · Jan – Apr 2026 · GitHub · Article · Live

- Peer review as a coordination game: authors submit research artifacts pinned to IPFS and stake ETH, a LangGraph submission agent screens plagiarism (Exa AI + Gemini), Chainlink VRF v2.5 selects reviewers from the staked pool, and human verdicts settle on-chain through a LangGraph review agent (HITL, EIP-712 signatures, on-chain transactions).
- Getter-less contract with HMAC-verified Alchemy webhooks projecting events into Neon Postgres (optimistic concurrency); Inngest + LangGraph checkpointers carry multi-day human review across cold starts. 50 Foundry tests; Base Sepolia + Ethereum Sepolia.

More projects

- **Bet2Gether** · Oct – Nov 2025 · GitHub · Live — Chainlink-settled prediction market on Ethereum Sepolia: Automation resolves rounds from Price Feeds, VRF v2.5 picks ERC-1155 rewards, time-weighted payouts (oracle staleness limitation documented).
- **Forge** · Nov 2025 · GitHub · Live — atomic ERC-1155 crafting (burnBatch + mint in one transaction), rules contract separate from the ledger, 37 Foundry tests at 100% coverage.
- **GavL** · Sep – Oct 2025 · GitHub · Live — Next.js + Supabase real-time auctions: Realtime bids, CRON-triggered closure, Stripe webhooks with an RLS-safe service role, i18n FR / EN / DE.

EARLIER CAREER (PRE-SOFTWARE)

KIBAG Gruppe · SERPOL · SITA Remediation (SUEZ) — France & Switzerland · Land Remediation Engineer / Project Manager · 2012 – 2020 — Led environmental fieldwork, soil-sampling operations, subcontractor coordination, and compliance reporting for industrial remediation sites.

University of British Columbia · INSERM — Canada & France · Molecular Biology Researcher · 2001 – 2009 — Genomic research in neural stem cells and epigenetics; co-author on three peer-reviewed publications (J. Neurosci. Res. · Fungal Genet. Biol. · Dev. Biol.).

EDUCATION

2025 — EVM Architecture & Smart Contract Security (core curriculum) — Metana (remote)

2021 – 2023 — M.Sc. Blockchain Technologies — University of Barcelona (remote)

2022 — Full-Stack Software Engineering Bootcamp — Le Wagon, Zurich

2012 — M.Eng. Civil & Environmental Engineering, École des Mines · 2001 — M.Sc. Molecular Biology & Genetics, Univ. of Montpellier (France)